

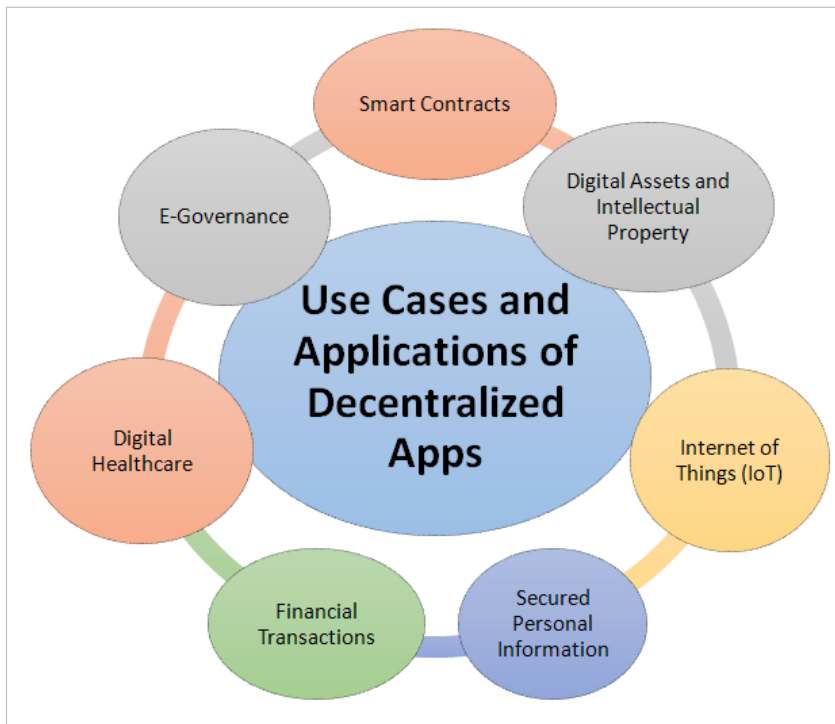


Figure 1: Use cases and applications of decentralised apps

fraud and streamlining the process of transferring property titles. Blockchain-powered platforms are also emerging, enabling fractional ownership of real estate and expanding investment opportunities to a broader audience.

The energy sector is also benefiting from blockchain technology. By leveraging blockchain, decentralised energy trading platforms can be established, allowing peer-to-peer energy transactions and promoting the use of renewable energy sources. Blockchain-based systems can track energy production and consumption, facilitating efficient energy management and ensuring a reliable supply.

These examples illustrate the real-world applications of blockchain technology. From finance, supply chain management, and healthcare to real estate and energy, blockchain's decentralised and transparent nature opens up vast possibilities for innovation. As the technology continues to evolve and mature, it is expected to find even more

diverse applications, transforming various industries and driving positive change in how we interact and conduct business.

The role of databases in the blockchain environment

Databases play a crucial role in supporting the functioning and efficiency of blockchain environments. They provide a means to store and manage data related to blockchain transactions, smart contracts, and other associated information.

One important use of databases in a blockchain environment is to store off-chain data. Blockchain networks typically prioritise on-chain data storage to maintain the integrity and immutability of transaction records. However, there are instances where it may not be practical or efficient to store certain data, such as large files, multimedia content, or extensive metadata, directly on the blockchain. In such cases, databases are used to store this off-chain data while referencing it through cryptographic hashes or

pointers on the blockchain. For example, IPFS (InterPlanetary File System) is a distributed database that is often used in conjunction with blockchain to store and retrieve large files, ensuring efficient storage and retrieval of data while maintaining a link to the blockchain.

Another important role of databases in blockchain environments is to provide indexing and querying capabilities. Blockchain networks are designed for secure and decentralised transaction storage, but they are not optimised for complex data retrieval and querying. Databases can be used to create indexes and data structures that allow for efficient search and retrieval of specific information from the blockchain. By using databases in conjunction with blockchain, developers and users can leverage the benefits of both technologies. For instance, Ethereum, a popular blockchain platform, uses a database called LevelDB to index and query its transaction data, enabling faster and more efficient data retrieval.

These databases are used to store and manage the state of smart contracts in blockchain environments. Smart contracts are self-executing agreements that reside on the blockchain and automatically enforce the terms and conditions defined within them. As smart contracts can contain complex logic and state variables, databases provide a way to store and update the current state of these contracts. For example, in Ethereum, the state of smart contracts is stored in a database called the 'state trie', which allows for efficient contract execution and retrieval of contract states.

Databases are also employed in blockchain environments to support scalability and performance. While blockchain networks aim to achieve decentralised consensus and immutability, they often face challenges in terms of transaction throughput and latency. By using databases that are designed for high-performance data storage and retrieval, blockchain systems can offload certain